

Locating the Prime Carrier: the ζ Prime Signal is the Value-Space Pair Correlation of the Zeros

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Companion to *Spectral Rigidity and Shuffled Spacing Discriminant* — SIP License v1.1

Abstract

The shuffle-knife discriminant established that, among the standard spectral witnesses of the Riemann zeros, the arithmetic prime-resonance is the one carrying ζ -specific structure (FUNCTION, $\sim 11,500\sigma$) while the distributional witnesses read only the universality class (FORM). It located the signal by *elimination*. This note locates it by *construction*: it identifies, positively and to numerical precision, the single two-point object the prime signal lives in. The prime-resonance power $\mathcal{P} = \sum_p |\sum_j e^{-i(\log p)\gamma_j}|^2/N$ is, by expansion, a linear functional of the empirical value-space pair correlation of the zero *positions*; reconstructing $\mathcal{P}$ from that pair correlation alone recovers 100.0% of the signal (exact to machine precision), whereas surrogates that preserve the two-point statistics of the *spacings* or the index-space fluctuations recover only $\sim 7\%$. A fine sweep of the position form factor peaks only at frequencies $\log n$ with n a prime power, with heights tracking the explicit-formula weight $(\Lambda(n)/\sqrt{n})^2$ at Pearson $r = 0.9975$ and vanishing (~ 0.01) at composite n . The separation from the shuffled floor grows from $18\times$ at $N = 200$ to $474\times$ at $N = 10^4$. The operational conclusion is that “the primes are not a two-point statistic” — a claim we had entertained internally from spacing-domain surrogate tests — is false and is retracted here: the primes are exactly the two-point pair correlation of the positions, i.e. the Montgomery form factor read through the Riemann–von Mangoldt explicit formula. The word “two-point” is simply ambiguous between the positions and the spacings; the carrier is the former, and the surrogate collapse measures the destruction of that object, not the presence of any higher-order one.

1 The ambiguity that motivated the test

The prime-resonance witness of the shuffle-knife study is a scalar built from the actual zero heights,

$$\mathcal{P} = \sum_p P(\log p), \quad P(f) = \frac{1}{N} \left| \sum_{j=1}^N e^{-if\gamma_j} \right|^2, \quad (1)$$

summed here over $p \in \{2, 3, 5, 7, 11, 13\}$, with $\gamma_1 < \dots < \gamma_N$ the ordinates of the nontrivial zeros. The knife showed that $\mathcal{P}$ collapses under a marginal-matched permutation of the spacings $s_i = \gamma_{i+1} - \gamma_i$: the signal is destroyed when the *order* of the spacings is destroyed. A natural but unguarded reading of that collapse — one we recorded in an internal strip-mine synthesis — is that the prime signal is inaccessible to any two-point statistic and must reside in higher-order phase coherence, so that recovering it (the “converse machine” of the analogous Fan–Wan lesson) would require a genuinely new, higher-order instrument.

That reading conflates two different two-point objects. Expanding (1),

$$P(f) = \frac{1}{N} \sum_{j,l} e^{-if(\gamma_j - \gamma_l)} = \int e^{-ifs} \rho_2(s) ds, \quad \rho_2(s) = \frac{1}{N} \sum_{j,l} \delta(s - (\gamma_j - \gamma_l)), \quad (2)$$

so $P(f)$ is by construction the Fourier transform of ρ_2 , the empirical *value-space pair correlation of the positions*. It is a two-point statistic — of the positions. The shuffle knife destroys a *different* two-point object, the correlation of the spacing sequence, which does not determine ρ_2 because the map spacings $\mapsto$ positions is a cumulative sum. Whether the primes are “two-point” therefore has no answer until one says two-point *of what*. This note settles it by direct measurement.

Proposition 1 (Pair-correlation sufficiency). *$\mathcal{P}$ is a linear functional of ρ_2 alone; hence any transformation of the point set that preserves the value-space pair correlation of the positions preserves $\mathcal{P}$ exactly.*

This is immediate from (2); the content of the paper is the measurement of *which* surrogate operations preserve ρ_2 and which annihilate it, and the identification of the peak structure with the explicit formula.

2 Method

All measurements reuse the functional (1) verbatim from the shuffle-knife suite (canonical seed 20260423) and the first N ordinates of the Riemann zeros. We evaluate the following, each a distinct answer to “two-point of what?”:

Baseline. $\mathcal{P}$ on the ordered zeros.

Spacing floor. $\mathcal{P}$ after a random permutation of $\{s_i\}$ (one-point spacing distribution only). Averaged over 300 permutations.

Gap two-point (C). $\mathcal{P}$ after phase randomisation of the spacing series, which *preserves the spacing autocovariance* (the two-point statistic of the gaps) and destroys all higher-order gap structure.

Index two-point (D). $\mathcal{P}$ after phase randomisation of the position fluctuation about a smooth degree-5 counting trend, preserving the index-space autocovariance of the positions.

Value-space pair correlation (E). $\mathcal{P}$ recomputed from ρ_2 via (2) (exact, and a 4000-bin histogram variant).

Phase randomisation preserves a power spectrum, hence exactly a two-point autocovariance, while destroying phase relations; it is the standard surrogate for isolating what a two-point statistic can and cannot carry. As calibration we require (i) the ρ_2 reconstruction to reproduce the baseline to machine precision, and (ii) a Poisson point process of matched mean density to show no prime peak above the floor.

3 Results

Which two-point object carries the signal (N=200, 300 trials).

object preserved	$\mathcal{P}$	% baseline
ordered baseline	51.99	100.0%
spacing shuffle (one-point)	2.77	5.3%
gap two-point (C, spacing autocovariance)	3.77	7.3%
index two-point (D, position-fluctuation spectrum)	3.73	7.2%
value-space pair correlation (E, exact)	51.99	100.0%
value-space pair correlation (E, 4000-bin)	51.66	99.4%
Poisson null (calibration)	5.87	11.3%

The pair-correlation reconstruction reproduces the baseline to $< 10^{-9}$ (calibration passed); the coarse 4000-bin histogram still recovers 99.4%. The two spacing/index two-point surrogates recover essentially nothing. *The primes are the value-space pair correlation of the positions; they are absent from the gap and index two-point objects.*

The peaks are the explicit formula (N=4000). A fine sweep of the position form factor $K(f) = |\sum_j e^{-if\gamma_j}|^2/N$ over $f \in (0.5, \log 40)$ shows peaks *only* at $f = \log n$ with n a prime power, and silence at composite n . Measured peak heights track the Riemann-von Mangoldt weight $\Lambda(n)/\sqrt{n}$, where Λ is the von Mangoldt function ($\log p$ at prime powers $n = p^k$, else 0):

n	type	measured height	$\Lambda(n)/\sqrt{n}$
5	prime	66.0	0.720
7	prime	69.3	0.736
11	prime	64.8	0.723
13	prime	61.3	0.711
4	prime power 2^2	14.1	0.347
8	prime power 2^3	6.9	0.245
9	prime power 3^2	15.4	0.366
16	prime power 2^4	3.7	0.173
32	prime power 2^5	1.6	0.123
6, 10, 12, 15	composite	~ 0.00	0

Mean height at prime-power frequencies is 36.6 against 0.01 at composite frequencies. The correlation between measured prime-power peak height and $(\Lambda(n)/\sqrt{n})^2$ is

$$\text{Pearson } r = 0.9975.$$

Heights fall with n exactly as $1/\sqrt{p^k}$ (so 4, 8, 9, 16, 32 shrink monotonically) and vanish where $\Lambda = 0$ (composites). This is the explicit-formula coupling between zeros and prime powers, measured directly on the form factor; it also disposes of the collateral internal conjecture that the two-point form factor $K(\tau)$ would fail to show the primes.

Robustness in N .

N	baseline $\mathcal{P}$	shuffle floor	separation
200	51.99	2.83	18.4×
500	88.68	2.13	41.7×
1000	137.53	1.87	73.6×
2000	216.40	1.75	123.4×
5000	407.69	1.53	266.9×
10000	672.90	1.42	474.2×

The value-space reconstruction equals the baseline at every N (Proposition 1); the separation from the floor (each floor averaged over 200 spacing permutations) sharpens with N , consistent with the shuffle-knife amplification result. The floor is a small, mildly noisy quantity; the separations are reproducible from `scripts/prime_carrier_reproduce.py` at the canonical seed but should be read as $\mathcal{O}(10)$ -to- $\mathcal{O}(10^2)$ growth, not to the last digit.

Coherence and mechanism. The raw peak $|\sum_j e^{-i(\log 2)\gamma_j}|^2$ grows faster than N ($|\cdot|^2/N$: $8.1 \rightarrow 50.5$ across $N = 500 \rightarrow 8000$; $|\cdot|^2/N^2$: $0.016 \rightarrow 0.006$), i.e. the peak is a coherent, non-self-averaging arithmetic sum. And the gap two-point surrogate is *faithful*: the spacing autocovariance at lags 1, 2, 3 is $(-0.009, 0.061, 0.067)$ on the object and $(-0.009, 0.061, 0.067)$ on the surrogate, yet $\mathcal{P}$ falls from 216.4 to 1.5. A surrogate that provably preserves the gap two-point object still loses the primes — the carrier is the position pair correlation, not the gap correlation.

4 What this establishes, and the honest edges

Establishes. The ζ prime-resonance signal is, positively and to machine precision, the value-space pair correlation of the zero *positions* — equivalently the single-realisation spectral form factor — whose peaks are the prime powers with explicit-formula weights ($r = 0.9975$). The prior internal reading “no two-point statistic carries the primes; a higher-order machine is required” is **retracted**: it measured the wrong two-point object (the spacings), and the surrogate collapse it observed is the destruction of ρ_2 , not the signature of higher-order structure. The “converse machine” is therefore not new: it is the pair correlation / explicit formula already central to the Montgomery–Odlyzko picture.

Nuance, not overclaimed. The signal lives in the *single-realisation* form factor of the actual zeros. The ensemble-averaged pair correlation tends to the smooth GUE law; the prime peaks are the arithmetic, non-self-averaging part (the off-diagonal/diagonal decomposition of the form factor). “Two-point carries the primes” is a statement about the empirical form factor of the real zeros, not about the universal GUE R_2 . This is consistent with the shuffle knife: the distributional witnesses are FORM because they see only the universal part; the arithmetic witness is FUNCTION because it sees the non-universal peaks that *are* ρ_2 ’s deviation from GUE.

Where the method stops. The identification is spectral and requires enough resolved zeros for the peaks to stand above the floor. For an L -function with only a handful of reachable zeros (e.g. the elliptic curve 11a1, with ~ 10 ordinates in the public tables) the form factor cannot resolve its arithmetic peaks at reachable height. That data scarcity is exactly the measurability wall the signal-to-noise criterion names elsewhere in this programme: the machine is known and universal, but the spectral data must exist to run it. We mark this boundary so the positive result is not over-read as a claim about objects whose spectra we cannot yet reach.

Companion to *Spectral Rigidity and Shuffled Spacing Discriminant* and to *Riemann Spectral Critical Line*. All measurements use the real first N Riemann zeros and the canonical seed 20260423; the functional (1) is the shuffle-knife prime-resonance witness reused verbatim, and the reconstruction/sweep are reproducible standalone. Nothing here proves the Riemann Hypothesis or constructs a Hilbert–Pólya operator; “carrier” means “the two-point object the prime-resonance functional is a functional of,” identified and calibrated. SIP License v1.1.